

Data Structures

CSE-207

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Today's topics



Introducing

Syllabus

Book

Exam

About Course



- Course Code: **CSE-207**
- Course Title: **Data Structures**
- Course Teacher: **Md. Manowarul Islam**
- Credit: **4**

Overview



- Number of Class: 24
- Number of Mid-Term : 2
- Number of Quiz : 3
- Final Exam: 1

These may
vary !!!!!

Overview

- Number of Class: 24
- Number of Mid-Term : $130*1=30$
- Number of Quiz : CT $10=10$
- Final Exam: 1 $30*1=30$
- Lab : $=15$
- Project $=10$
- Attendance $=5$

Syllabus



Introduction

- ❖ Elementary data organization
- ❖ Data structures
- ❖ Data structure operations
- ❖ Algorithmic notation
- ❖ String processing

Syllabus

Arrays

- ❖ Linear arrays
- ❖ Representation of array
- ❖ Traversing linear arrays
- ❖ Inserting and deleting in arrays
- ❖ Bubble sort
- ❖ Linear search
- ❖ Binary search
- ❖ Multidimensional arrays

Array:

23	4	6	15	5	7
0	1	2	3	4	5

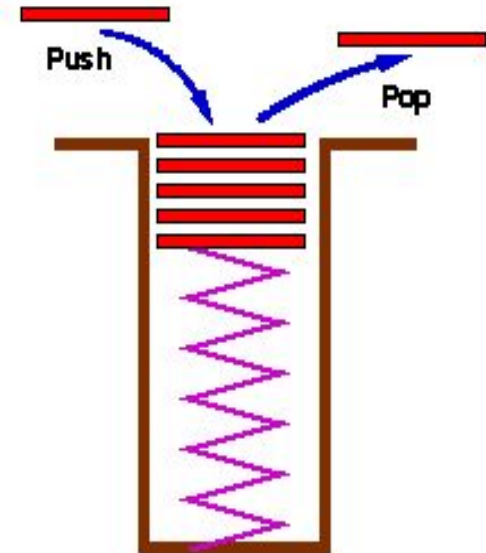


Array index

Syllabus

Stack

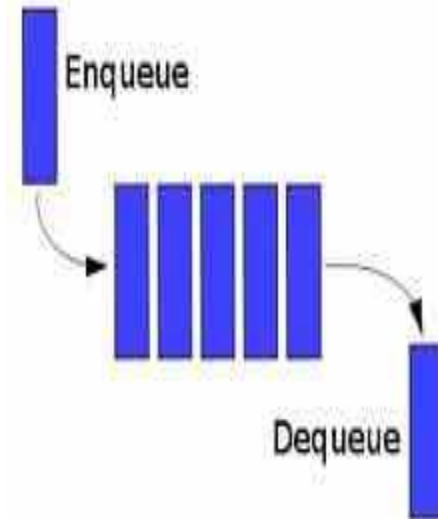
- ❖ Stack
- ❖ Array representation of stack
- ❖ Linked representation of stack
- ❖ Polish notation



Syllabus

Queues

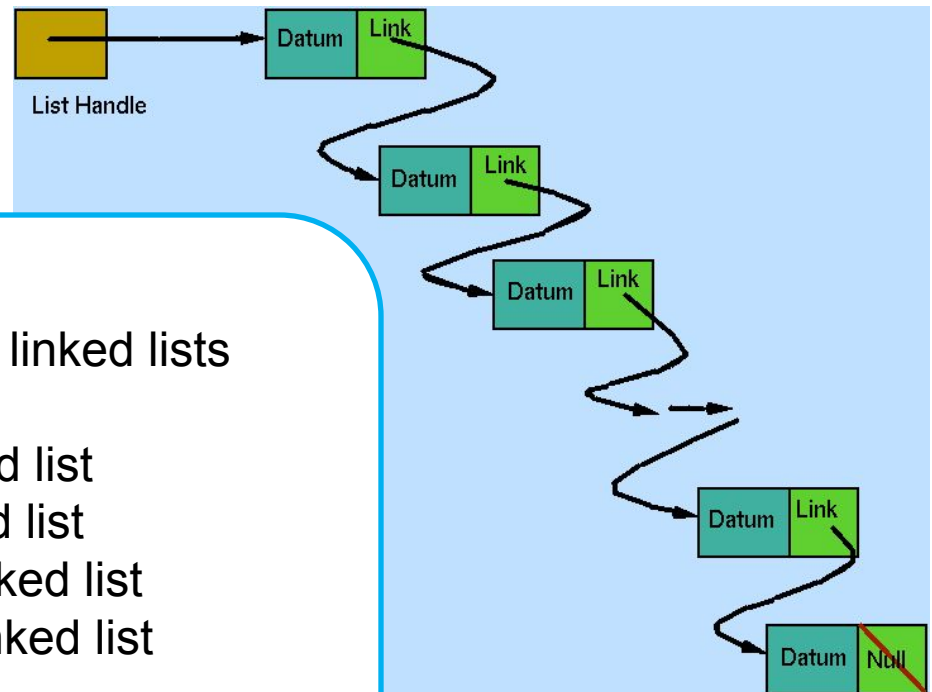
- ❖ Queues
- ❖ Array representation of queue
- ❖ Deques
- ❖ Priority queue



Syllabus

Linked Lists

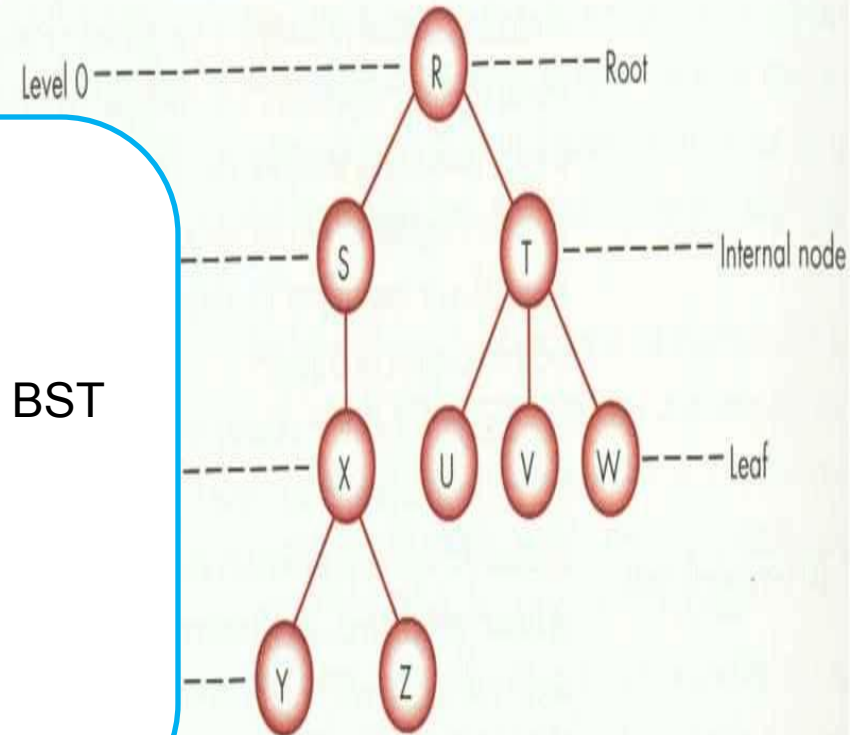
- ❖ Linked lists
- ❖ Representation of linked lists in memory
- ❖ Traversing a linked list
- ❖ Searching a linked list
- ❖ Insertion into a linked list
- ❖ Deletion from a linked list
- ❖ Header linked list
- ❖ Two way lists



Syllabus

Trees

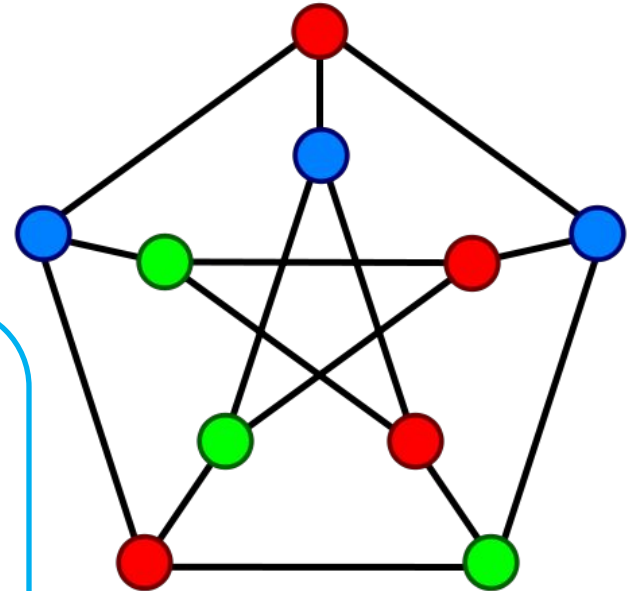
- ❖ Binary trees
- ❖ Representing binary trees
- ❖ Traversing binary trees
- ❖ Binary search trees
- ❖ Searching and inserting in BST
- ❖ Deleting in BST
- ❖ Heap
- ❖ Heap sort
- ❖ Huffman's algorithm



Syllabus

Graphs

- ❖ Graph
- ❖ Adjacency matrix
- ❖ Path matrix
- ❖ Warshall's algorithm
- ❖ Linked representation of graph
- ❖ BFS
- ❖ DFS



Syllabus

Sorting

- ❖ Insertion sort
- ❖ Selection sort
- ❖ Merge Sort
- ❖ Bubble sort



Syllabus

Searching

- ❖ Searching technique
- ❖ Linear search
- ❖ Binary search



Exam



Book

❖ *Data Structures*

Seymour Lipschutz

❖ *Introduction to Algorithm*

T H. Cormen

C. E. Leiserson

R. L Rivest

C. Stein



Any Question???